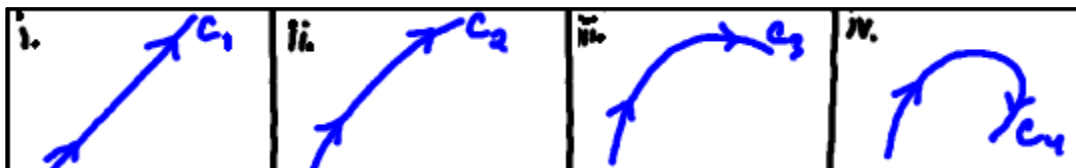


## 2.5 Curvature

We will now explore the notion of curvature. We begin by reviewing some terminology.

**Definition:** A parameterization  $\mathbf{r}(t)$  is called smooth on an interval  $I$  if  $\mathbf{r}'$  is continuous and  $\mathbf{r}'(t) \neq \mathbf{0}$ . A curve is smooth if it has a smooth parameterization. A smooth curve has no sharp corners or cusps.

**Remark:** For a smooth curve given by  $\mathbf{r}(t)$  when the unit tangent vector,  $\mathbf{T}(t)$ , turns it does so continuously. When a smooth curve has sharp turns  $\mathbf{T}(t)$  will turn quickly. When a curve turns quite slowly,  $\mathbf{T}(t)$  will not turn quickly. Consider the pictures below for examples of this idea.



This leads to the following definition.

**Definition:** The curvature of a smooth curve is  $\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|$ .

### How does the definition work?

If we can find the arc length parameterization for  $\mathbf{r}(t(s))$ , we can get  $\mathbf{T}(s)$  and then calculate  $\|\mathbf{T}'(s)\|$ .

Suppose that we have the vector function  $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$  and what to find the curvature here.

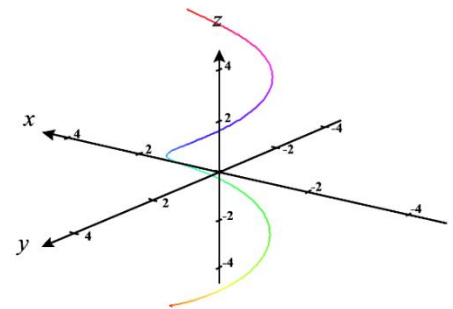
We first note that  $\mathbf{r}'(t) = \langle -\sin(t), \cos(t), 1 \rangle$ . Clearly  $\mathbf{r}'(t)$  is continuous and  $\mathbf{r}'(t) \neq \mathbf{0}$  for any  $t$ , so we can say that  $\mathbf{r}(t)$  is smooth.

Therefore we define the arc length function of  $\mathbf{r}(t)$  starting from  $t = 0$  to be:

$$s(t) = \int_0^t \|\mathbf{r}'(u)\| du = \int_0^t \sqrt{2} du = \sqrt{2}t$$

This implies that  $t = \frac{s}{\sqrt{2}}$ .

Therefore, we have an arc length parameterization of  $\mathbf{r}(t(s)) = \left\langle \cos\left(\frac{s}{\sqrt{2}}\right), \sin\left(\frac{s}{\sqrt{2}}\right), \frac{s}{\sqrt{2}} \right\rangle$ .



Now we attempt to calculate  $\mathbf{T}(s) = \frac{\mathbf{r}'(s)}{\|\mathbf{r}'(s)\|}$ .

We see that  $\mathbf{r}'(s) = \left\langle -\frac{1}{\sqrt{2}} \sin\left(\frac{s}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \cos\left(\frac{s}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \right\rangle$  and

$$\|\mathbf{r}'(s)\| = \sqrt{\frac{1}{2} \sin^2\left(\frac{s}{\sqrt{2}}\right) + \frac{1}{2} \cos^2\left(\frac{s}{\sqrt{2}}\right) + \frac{1}{2}} = \sqrt{1} = 1.$$

$$\text{Thus } \mathbf{T}(s) = \frac{\mathbf{r}'(s)}{\|\mathbf{r}'(s)\|} = \left\langle -\frac{1}{\sqrt{2}} \sin\left(\frac{s}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \cos\left(\frac{s}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \right\rangle.$$

Therefore,  $\frac{d\mathbf{T}}{ds} = \mathbf{T}'(s) = \left\langle -\frac{1}{2} \cos\left(\frac{s}{\sqrt{2}}\right), -\frac{1}{2} \sin\left(\frac{s}{\sqrt{2}}\right), 0 \right\rangle$  and thus

$$\left\| \frac{d\mathbf{T}}{ds} \right\| = \sqrt{\frac{1}{4} \cos^2\left(\frac{s}{\sqrt{2}}\right) + \frac{1}{4} \sin^2\left(\frac{s}{\sqrt{2}}\right) + 0} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

So we say that the curvature for  $\mathbf{r}(t)$  is  $\kappa = \frac{1}{2}$ .

**Theorem:** The curvature of a smooth curve given by  $\mathbf{r}(t)$  can be expressed as

$$1) \quad \kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}.$$

OR

$$2) \quad \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

*Proof:*

(1) First, note that by the chain rule that  $\frac{d\mathbf{T}}{dt} = \frac{d\mathbf{T}}{ds} \cdot \frac{ds}{dt}$ . By the Fundamental Theorem of Calculus we know

that  $\frac{ds}{dt} = \|\mathbf{r}'(t)\|$ . Since the curve is smooth we know that  $\|\mathbf{r}'(t)\| \neq 0$ . This means

$$\frac{d\mathbf{T}}{dt} = \frac{d\mathbf{T}}{ds} \cdot \frac{ds}{dt} \Rightarrow \frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}/dt}{ds/dt} \Rightarrow \frac{d\mathbf{T}}{ds} = \frac{\mathbf{T}'(t)}{\|\mathbf{r}'(t)\|}. \text{ Finally, we have that } \left\| \frac{d\mathbf{T}}{ds} \right\| = \left\| \frac{\mathbf{T}'(t)}{\|\mathbf{r}'(t)\|} \right\| = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$$

which means that  $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$  as desired. ■

(2) We know, by definition, that

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} \Rightarrow \mathbf{r}'(t) = \|\mathbf{r}'(t)\| \mathbf{T}(t) \Rightarrow \mathbf{r}''(t) = \|\mathbf{r}'(t)\|' \mathbf{T}(t) + \|\mathbf{r}'(t)\| \mathbf{T}'(t).$$

We also know, by definition, that  $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} \Rightarrow \mathbf{T}'(t) = \|\mathbf{T}'(t)\| \mathbf{N}(t)$  and by (a) we know that

$$\kappa = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} \Rightarrow \|\mathbf{T}'(t)\| = \kappa \|\mathbf{r}'(t)\| \text{ therefore we can see that } \mathbf{T}'(t) = \kappa \|\mathbf{r}'(t)\| \mathbf{N}(t).$$

If we use this to substitute into  $\mathbf{r}''(t) = \|\mathbf{r}'(t)\| \mathbf{T}'(t) + \|\mathbf{r}'(t)\|' \mathbf{T}(t)$  we obtain that

$$\mathbf{r}''(t) = \|\mathbf{r}'(t)\| \kappa \|\mathbf{r}'(t)\| \mathbf{N}(t) + \|\mathbf{r}'(t)\|' \mathbf{T}(t) = \kappa \|\mathbf{r}'(t)\|^2 \mathbf{N}(t) + \|\mathbf{r}'(t)\|' \mathbf{T}(t).$$

If we now compute  $\mathbf{r}'(t) \times \mathbf{r}''(t)$  we obtain:

$$\begin{aligned} \mathbf{r}'(t) \times \mathbf{r}''(t) &= \left[ \|\mathbf{r}'(t)\| \mathbf{T}(t) \right] \times \left[ \kappa \|\mathbf{r}'(t)\|^2 \mathbf{N}(t) + \|\mathbf{r}'(t)\|' \mathbf{T}(t) \right] \\ &= \kappa \|\mathbf{r}'(t)\|^3 [\mathbf{T}(t) \times \mathbf{N}(t)] + \|\mathbf{r}'(t)\| \|\mathbf{r}'(t)\|' [\mathbf{T}(t) \times \mathbf{T}(t)] = \kappa \|\mathbf{r}'(t)\|^3 \mathbf{B}(t) \end{aligned}$$

However, since  $\mathbf{B}(t)$  is a unit vector it follows that  $\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \kappa \|\mathbf{r}'(t)\|^3$ .

Therefore, we have that  $\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$

■

*Note: These formulas provide us with a much easier way to compute the curvature compared with computing directly from the definition.*

*Note: Part (b) of this theorem will tend to be more useful since it only requires that we know the vector-valued function  $\mathbf{r}(t)$ , we don't need to compute the tangent vector  $\mathbf{T}(t)$ , and we can easily change the parameter if needed.*

**Example 1:** Use each of the formulas from the theorem above to calculate the curvature of  $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ .

Using  $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$ :

We saw above that  $\mathbf{r}'(t) = \langle -\sin(t), \cos(t), 1 \rangle$  and so we note that  $\|\mathbf{r}'(t)\| = \sqrt{2}$  and therefore,

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \left\langle -\frac{1}{\sqrt{2}} \sin(t), \frac{1}{\sqrt{2}} \cos(t), \frac{1}{\sqrt{2}} \right\rangle.$$

We then calculate that  $\mathbf{T}'(t) = \left\langle -\frac{1}{\sqrt{2}} \cos(t), -\frac{1}{\sqrt{2}} \sin(t), 0 \right\rangle$  and

$$\|\mathbf{T}'(t)\| = \sqrt{\frac{1}{2} \cos^2(t) + \frac{1}{2} \sin^2(t) + 0} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

Therefore,  $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\left(\frac{1}{\sqrt{2}}\right)}{\sqrt{2}} = \frac{1}{2}.$

Using  $\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$ .

We know that that  $\mathbf{r}'(t) = \langle -\sin(t), \cos(t), 1 \rangle$  and so  $\mathbf{r}''(t) = \langle -\cos(t), -\sin(t), 0 \rangle$ .

We calculate that  $\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\sin(t) & \cos(t) & 1 \\ -\cos(t) & -\sin(t) & 0 \end{vmatrix} = \langle \sin(t), \cos(t), \sin^2(t) + \cos^2(t) \rangle = \langle \sin(t), \cos(t), 1 \rangle$

Note that  $\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \sqrt{2}$  &  $\|\mathbf{r}'(t)\| = \sqrt{2}$ .

Therefore,  $\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3} = \frac{\sqrt{2}}{(\sqrt{2})^3} = \frac{1}{2}$ .

**Example 2:** Consider the vector function  $\mathbf{r}(t) = \left\langle t, t^2, \frac{2}{3}t^3 \right\rangle$ .

a) Use any approach to calculate the curvature of  $\mathbf{r}(t)$ .

We note that  $\mathbf{r}'(t) = \langle 1, 2t, 2t^2 \rangle$  and  $\mathbf{r}''(t) = \langle 0, 2, 4t \rangle$ .

We calculate that  $\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2t & 2t^2 \\ 0 & 2 & 4t \end{vmatrix} = \langle 8t^2 - 4t^2, -(4t - 0), 2 - 0 \rangle = \langle 4t^2, -4t, 2 \rangle$

Note that

$$\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \sqrt{16t^4 + 16t^2 + 4} = \sqrt{4(2t^2 + 1)^2} = 2(2t^2 + 1) \text{ \& } \|\mathbf{r}'(t)\| = \sqrt{1 + 4t^2 + 4t^4} = \sqrt{(2t^2 + 1)^2} = (2t^2 + 1).$$

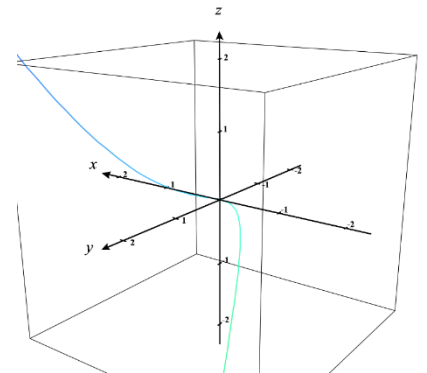
Therefore,  $\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3} = \frac{2(2t^2 + 1)}{(2t^2 + 1)^3} = \frac{2}{(2t^2 + 1)^2}$ .

*Note: The curvature formula here depends on the value of  $t$ ! This means that the curve  $\mathbf{r}(t)$  isn't equally "bendy" at all points. If we evaluate this formula at a specific value of  $t$  we can determine the curvature at that time on the curve.*

b) Calculate the curvature when  $t = 0, t = 1$ , &  $t = 2$ .

We see that

$$\kappa(0) = \frac{2}{(2t^2 + 1)^2} = 2 \quad \kappa(1) = \frac{2}{25} \quad \kappa(2) = \frac{2}{81}$$



*Note: We can visually see the different amounts of "bend" on the curve itself and note that the curve appears the most "bent" when  $t = 0$  (meaning we are at the point  $(0,0,0)$ ).*

**Example 3:** Consider the circle given by  $\mathbf{r}(t) = \langle a \cos(t), a \sin(t) \rangle$  where  $0 \leq t < 2\pi$  and  $a > 0$ . Find the curvature of this circle.

We note that  $\mathbf{r}'(t) = \langle -a \sin(t), a \cos(t) \rangle$  and  $\mathbf{r}''(t) = \langle -a \cos(t), -a \sin(t) \rangle$ . So we calculate that:

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -a \sin(t) & a \cos(t) & 0 \\ -a \cos(t) & -a \sin(t) & 0 \end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} + a^2\mathbf{k} \quad \text{and}$$

$$\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \sqrt{0 + 0 + a^4} = a^2$$

$$\|\mathbf{r}'(t)\|^3 = \left( \sqrt{a^2 \sin^2(t) + a^2 \cos^2(t)} \right)^3 = \left( \sqrt{a^2} \right)^3 = a^3$$

$$\text{Therefore, } \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3} = \frac{a^2}{a^3} = \frac{1}{a}$$

OR (using (1) from the theorem to calculate the curvature)

We note that  $\mathbf{r}'(t) = \langle -a \sin(t), a \cos(t) \rangle$ . This means

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \frac{1}{\sqrt{a^2 \sin^2(t) + a^2 \cos^2(t)}} \langle -a \sin(t), a \cos(t) \rangle = \frac{1}{a} \langle -a \sin(t), a \cos(t) \rangle = \langle -\sin(t), \cos(t) \rangle. \text{ We}$$

note from the above calculation that  $\|\mathbf{r}'(t)\| = a$  and  $\mathbf{T}'(t) = \langle -\cos(t), -\sin(t) \rangle$ . So, by the above theorem the

$$\text{curvature is } \kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\|\langle -\cos(t), -\sin(t) \rangle\|}{a} = \frac{1}{a}.$$

*Note: Intuitively this means that if a circle has a small radius, its curvature is high. This makes sense because over a fixed amount of arc length small circles turn more abruptly than large circles.*

*Note: This example gives us a way to visualize the amount of “bend” that a given curvature looks like. For example, if we calculate, as in Example 1, that we have  $\kappa = \frac{1}{2}$  then we know that this is the same amount of “bend” that we would see in a circle of radius 2.*

**Example 4:** Let  $C$  be the ellipse  $x^2 + 16y^2 = 16$  oriented counterclockwise. Measure the curvature at each endpoint of its major and minor axes.



We note that  $\mathbf{r}(t) = 4\cos(t)\mathbf{i} + \sin(t)\mathbf{j}$ .

So,  $\mathbf{r}'(t) = -4\sin(t)\mathbf{i} + \cos(t)\mathbf{j}$  and  $\mathbf{r}''(t) = -4\cos(t)\mathbf{i} - \sin(t)\mathbf{j}$

The endpoints on the positive  $x$  and  $y$ -axis occur at  $t = 0$  and  $t = \frac{\pi}{2}$ . The curvature at the points on the  $x$ -axis are the same and the curvature on the  $y$ -axis are the same.

At  $t = 0$ :

$$\mathbf{r}'(0) = 1\mathbf{j} \Rightarrow \|\mathbf{r}'(0)\|^3 = 1$$

$$\mathbf{r}''(0) = -4\mathbf{i}$$

$$\mathbf{r}'(0) \times \mathbf{r}''(0) = 4\mathbf{k} \Rightarrow \|\mathbf{r}'(0) \times \mathbf{r}''(0)\| = 4$$

$$\text{Therefore, } \kappa(0) = \frac{\|\mathbf{r}'(0) \times \mathbf{r}''(0)\|}{\|\mathbf{r}'(0)\|^3} = \frac{4}{1} = 4$$

At  $t = \frac{\pi}{2}$ :

$$\mathbf{r}'\left(\frac{\pi}{2}\right) = -4\mathbf{i} \Rightarrow \left\|\mathbf{r}'\left(\frac{\pi}{2}\right)\right\|^3 = 4^3 = 64$$

$$\mathbf{r}''\left(\frac{\pi}{2}\right) = -\mathbf{j}$$

$$\mathbf{r}'\left(\frac{\pi}{2}\right) \times \mathbf{r}''\left(\frac{\pi}{2}\right) = 4\mathbf{k} \Rightarrow \left\|\mathbf{r}'\left(\frac{\pi}{2}\right) \times \mathbf{r}''\left(\frac{\pi}{2}\right)\right\| = 4$$

$$\text{Therefore, } \kappa\left(\frac{\pi}{2}\right) = \frac{\left\|\mathbf{r}'\left(\frac{\pi}{2}\right) \times \mathbf{r}''\left(\frac{\pi}{2}\right)\right\|}{\left\|\mathbf{r}'\left(\frac{\pi}{2}\right)\right\|^3} = \frac{4}{64} = \frac{1}{16}$$

*Note: It is important to note here that calculating the general curvature formula would be quite messy and so it is advantageous to only calculate the curvature at the specific values of  $t$  desired here.*